

Similarity-Aware Generalization Framework for Zero-shot Cross-Domain Sequential Recommendation

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ABSTRACT

Zero-shot cross-domain sequential recommendation aims to transfer sequential knowledge from an interaction-rich source domain to previously unseen target domains without any target-domain interaction data in training. Recent large language model (LLM)-based methods have improved this setting by leveraging item-side semantic information. However, existing alignment strategies typically enforce uniform cross-domain attraction over all item pairs. Such non-selective alignment may distort the latent semantic structures and weaken the domain-specific discrimination, especially when semantically weakly related items are forced to align. To address this issue, we propose **SAGERec**, a similarity-aware generalization framework for zero-shot cross-domain sequential recommendation. SAGERec mainly comprises a *Similarity-Aware Inter-domain Compactness* (SIC) module that selectively aligns the semantically compatible cross-domain item pairs, and a *Domain-Adaptive Generalization* (DAG) module that dynamically adjusts the strength of semantic regularization according to the discrepancy between the source and target domains. In this way, the proposed framework facilitates a reliable transfer while reducing noisy cross-domain attraction. Extensive experiments on multiple real-world datasets and representative sequential recommendation backbones demonstrate that SAGERec consistently improves recommendation performance within backbone families and achieves the best overall results across zero-shot transfer settings. Further analyses indicate that, compared with existing semantic-based and generalization-based baselines, SAGERec yields more structured cross-domain alignment and a more favorable efficiency–effectiveness trade-off.

1. Introduction

Sequential recommendation aims to model users' dynamic preferences from historical interactions and has become a fundamental component of modern recommender systems [1, 2]. However, most existing sequential recommenders are developed under the assumption that sufficient interaction data are available within a specific domain for model training and adaptation [3, 4, 5, 6]. In practice, this assumption is frequently violated. For example, when models trained on interaction-rich domains are deployed to newly emerging services or privacy-constrained platforms, the target-domain interactions may be sparse or entirely unavailable. Therefore, a primary challenge is how to effectively transfer knowledge from a high-resource source domain to support recommendations in a low-resource target domain [7].

Cross-Domain Sequential Recommendation (CDSR) addresses the aforementioned issue by transferring preference signals across domains. Nevertheless, most existing CDSR methods still rely on overlapping users or items, or require target-domain interactions for model adaptation [7]. These requirements restrict their applicability in realistic deployment scenarios, which motivates a practical and challenging setting, namely *Zero-shot Cross-Domain Sequential Recommendation* (ZCDSR). In this setting, a model trained on an

interaction-rich source domain is directly transferred to a previously unseen target domain without using any target-domain interaction data for optimization, while only the item-side content is available for transfer [8]. This objective is particularly challenging for sequential recommendation as the model must generalize across domains with substantial semantic and behavioral discrepancies while preserving sequential dependency patterns learned from the source domain.

Recent advances in large language models (LLMs) present a new pathway for addressing this challenge, driven by their strong semantic understanding and generalization ability acquired through large-scale pretraining [9, 10]. By encoding the item-side metadata (e.g., titles, descriptions and attributes) into dense semantic representations, LLMs make it possible to construct a shared semantic space across domains even without user or item overlap. This capability is especially appealing in zero-shot settings, where the target-domain interactions are unavailable for training while the item content is accessible at deployment time. Motivated by this advantage, recent studies have begun to explore LLM-driven zero-shot recommendation and have shown that the pretrained semantic representations can substantially improve transferability to unseen domains [8, 11, 12].

However, existing semantic generalization methods typically approach the cross-domain transfer in a rigid and uniform manner, mainly manifested in two levels. At the item level, they impose a global alignment between the source and target semantic spaces without distinguishing the varying degrees of semantic compatibility among specific

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item pairs. Such non-selective alignment overlooks the fine-grained structure of cross-domain relations. Specifically, the highly compatible pairs provide reliable transfer signals while the weakly related pairs may introduce semantic noise and lead to negative transfer. At the domain level, existing approaches consistently apply a fixed regularization strategy. This assumes the trade-off between the recommendation accuracy and generalization strength remains constant across different target domains. Such a one-size-fits-all strategy fails to account for the substantial semantic gaps caused by the differences in catalog structure, content modality and linguistic style. These limitations highlight the critical need for a new ZCDSR framework. Rather than relying on indiscriminate alignment, an effective solution can dynamically adapt its transfer strength based on specific discrepancies at both the item and domain levels.

To address the above challenges, we propose **SAGERec**, a **Similarity-Aware GEneralization** framework for zero-shot cross-domain sequential recommendation, which systematically resolves the limitations of existing methods at both the item and domain levels. First, to address the item-level problem of non-selective alignment, we design a similarity-aware semantic generalization objective. Within this objective, a *Similarity-Aware Inter-domain Compactness* (SIC) term selectively aligns the semantically compatible cross-domain item pairs with positive semantic affinity in the LLM embedding space, while an intra-domain diversity preservation term counteracts over-compression and retains the fine-grained item discrimination within each domain. Second, to address the domain-level problem of static regularization across heterogeneous domain pairs, we introduce a *Domain-Adaptive Generalization* (DAG) mechanism. DAG dynamically modulates the strength of semantic regularization according to the estimated discrepancy between the source and target domains, allowing a strong transfer across semantically close domains while enforcing the conservative alignment for distant ones. Ultimately, SAGERec offers a plug-and-play solution. It can be seamlessly incorporated into existing LLM-based recommenders without the need to modify the backbone model or introduce significant overhead. Extensive experiments on three real-world datasets show that SAGERec consistently outperforms the semantic-based and generalization-based baselines in zero-shot transfer scenarios.

The main contributions of this work are summarized as follows:

- We propose a **Similarity-Aware GEneralization** framework for ZCDSR, named **SAGERec**, which enhances the LLM-based semantic-sequential transfer pipeline by incorporating a similarity-aware semantic regularization into the transfer process.
- We design a *Similarity-Aware Inter-domain Compactness* mechanism to address the issue of non-selective cross-domain alignment, thereby reducing the noisy attraction by preventing weakly related items from being overly aligned.

- We introduce a *Domain-Adaptive Generalization* mechanism to address the limitation of static regularization across heterogeneous source-target domain pairs, thereby enabling the transfer strength to adapt to varying degrees of semantic discrepancy across domains.
- Extensive experiments on three real-world datasets and representative sequential backbones verify the effectiveness, robustness and efficiency of SAGERec over the competitive baselines.

The remainder of this paper is organized as follows. Section 2 reviews the related work. Section 3 formulates the problem definition. Section 4 details the proposed SAGERec framework. Subsequently, Section 5 reports the experimental results and analysis. Finally, Section 6 concludes the paper and discusses the future directions.

2. Related Work

2.1. Cross-Domain Sequential Recommendation

Cross-domain sequential recommendation (CDSR) aims to improve recommendation quality by transferring sequential preference signals across domains, and has been widely studied as an effective solution to data sparsity and cold-start issues. Early methods such as π -Net [13] and PSJNet [14] mainly rely on explicit information sharing across domains, while later approaches such as C²DSR [15] further incorporate graph modeling and contrastive learning to capture richer cross-domain dependencies. More recent studies have sought to relax the assumptions of conventional CDSR. For example, AMID [16] points out that many existing CDSR methods still depend on restrictive conditions such as overlapping users or stable train-test distributions, which may not hold in realistic environments. ADR-Rec [17] further improves CDSR by disentangling domain-specific and cross-domain preferences and dynamically controlling cross-sequence information with a gating mechanism. A recent survey [7] also shows that CDSR is gradually evolving toward more open and realistic transfer settings. Nevertheless, most existing CDSR methods still require some form of cross-domain overlap or target-domain interaction signal for adaptation, and are therefore not well suited to the more challenging setting where the target domain is previously unseen and no interaction data are available during training.

2.2. Zero-Shot Recommendation

Zero-shot recommendation studies how to generalize recommendation models to unseen users, unseen items, or entirely new domains without conventional supervised adaptation. Early work such as ZESRec [18] explicitly formulates zero-shot recommendation without overlapping users or items, while MAIL [19] addresses zero-shot cold-start recommendation through aligned auxiliary representations. Another related line focuses on transferable or pre-trained recommenders for zero-shot generalization, such as Pre-trained Neural Recommenders [20] and PrepRec [21], which improve transferability through universal interaction statistics or pre-trained sequential representations. In

addition, prompt-oriented paradigms such as P5 [22] and SCRIPT [23] further extend this direction by leveraging pre-training and prompting strategies. More recently, TCLF [24] highlights that, in the absence of target-domain interaction supervision, item content becomes a crucial bridge for zero-shot transfer, and richer alignment signals beyond pure content similarity can better support preference transfer. Although these studies demonstrate the importance of transferable semantic representations, they are not specifically designed for zero-shot cross-domain sequential recommendation, nor do they explicitly address how to organize and align cross-domain semantics under sequential transfer.

2.3. LLM-based Recommendation

The rapid development of large language models (LLMs) has stimulated extensive research on LLM-based recommendation [9]. Existing approaches can be broadly divided into two paradigms: generative recommendation, where LLMs are used as recommenders or ranking agents, such as TALL-Rec [25], GenRec [26], LlamaRec [27], and zero-shot conversational recommendation [28]; and encoder-based recommendation, where LLMs provide semantic embeddings for downstream recommenders. The latter paradigm is more closely related to our work. Harte et al. [29] show that LLM-derived embeddings can improve sequential recommendation, while LLM-ESR [30] demonstrates that cached semantic features can enhance recommendation without additional inference overhead. Safar et al. [31] further show that LLM-generated semantic embeddings, combined with FAISS-based similarity search, can improve both recommendation quality and retrieval efficiency across domains, highlighting the practical value of high-quality semantic representations for downstream recommendation and retrieval tasks. Most closely related to our work, LLM-RecG [8] studies zero-shot cross-domain sequential recommendation and identifies domain semantic bias as a major obstacle to transfer. These studies confirm that LLM-derived semantic representations provide an effective foundation for zero-shot cross-domain recommendation. However, existing methods still largely rely on uniform semantic generalization and pay limited attention to transfer heterogeneity across semantically different item pairs and domain pairs. This limitation motivates our similarity-aware and domain-adaptive framework for more robust zero-shot cross-domain sequential transfer.

3. Problem Definition

We study zero-shot cross-domain sequential recommendation (ZCDSR), where a model is trained on a source domain and transferred to a previously unseen target domain without using any target-domain interaction data for parameter optimization. The core challenge is to learn transferable sequential knowledge from an interaction-rich source domain and generalize it to a target domain where only source-domain interaction supervision is available during training.

We denote the source domain by

$$D^s = (U^s, V^s, X^s, S^s), \quad (1)$$

where U^s and V^s denote the user set and item set, respectively, X^s denotes the textual metadata of source-domain items, and S^s denotes the set of source-domain user interaction sequences. For each user $u^s \in U^s$, the interaction sequence is written as

$$s_{u^s} = [v_{u^s,1}^s, v_{u^s,2}^s, \dots, v_{u^s,T_{u^s}}^s], \quad (2)$$

where $v_{u^s,i}^s \in V^s$ is the item interacted with at position i , and T_{u^s} denotes the sequence length of user u^s .

Similarly, the target domain is denoted by

$$D^t = (U^t, V^t, X^t, S^t), \quad (3)$$

where U^t , V^t , X^t , and S^t are defined analogously for the target domain. In addition to the current target domain, we consider a set of auxiliary domains, denoted by

$$D^a = \{D^{a_1}, D^{a_2}, \dots, D^{a_M}\}, \quad (4)$$

where each auxiliary domain is a non-source domain that is not selected as the evaluation target in the current zero-shot task. For notational convenience, we define the set of all non-source domains as

$$D^{ns} = \{D^t\} \cup D^a. \quad (5)$$

Following the zero-shot setting, all non-source domains are disjoint from the source domain in both users and items:

$$\forall D^q \in D^{ns}, \quad U^s \cap U^q = \emptyset, \quad V^s \cap V^q = \emptyset. \quad (6)$$

During training, only source-domain interaction sequences S^s are available for supervised recommendation learning, while for each non-source domain $D^q \in D^{ns}$, only item-side metadata X^q is assumed to be available. Accordingly, the model cannot rely on interaction-based adaptation in the target domain, but must instead learn transferable sequential preference patterns from the source domain and generalizable item semantics from item-side information across domains.

Formally, let $f_{\Theta}(s, v)$ denote a parameterized scoring function that maps an interaction sequence s and a candidate item v to a real-valued relevance score. For notational simplicity, let $X = \{X^{\delta}\}_{D^{\delta} \in \{D^s\} \cup D^{ns}}$ denote the collection of item-side metadata from all domains. The goal of ZCDSR is to learn the optimal parameters Θ^* by jointly optimizing a source-domain recommendation objective and a cross-domain generalization objective:

$$\Theta^* = \arg \min_{\Theta} \mathcal{L}_{\text{rec}}(\Theta; S^s, X) + \Omega(\Theta; X) \mathcal{L}_{\text{gen}}(\Theta; X), \quad (7)$$

where $\mathcal{L}_{\text{rec}}$ denotes the source-domain recommendation objective defined over training instances derived from S^s , $\mathcal{L}_{\text{gen}}$ denotes a cross-domain generalization objective induced by item-side information from all domains, and $\Omega(\Theta; X)$ denotes a balancing term that controls the contribution of cross-domain generalization. Here, $\mathcal{L}_{\text{rec}}$ and $\mathcal{L}_{\text{gen}}$ are task-level abstractions, whose concrete formulations will be specified in the method section.

At inference time, the historical interaction sequence of a target-domain user is assumed to be observable. The learned scoring function is then applied to target-domain next-item ranking:

$$f_{\Theta^*} : S^t \times V^t \rightarrow \mathbb{R}, \quad (8)$$

which assigns a relevance score $f_{\Theta^*}(s_{u^t}, v^t)$ to each candidate item $v^t \in V^t$ conditioned on the observed target-domain interaction history $s_{u^t} \in S^t$, and ranks candidate items accordingly.

4. Method

Under the zero-shot protocol defined in Section 3, the goal is to transfer sequential knowledge learned from an interaction-rich source domain to a target domain without using any target-domain interaction data for optimization. In our setting, supervised learning is performed only on source-domain interaction sequences, while for non-source domains, including the target domain and auxiliary domains, only item-side semantic metadata are accessible and used for interaction-free semantic regularization. To address this problem, we propose **SAGERec**, a **S**imilarity-Aware **G**eneralization framework for zero-shot cross-domain sequential recommendation.

As illustrated in Fig. 1, SAGERec builds on a standard LLM-enhanced semantic-sequential transfer pipeline and improves its robustness through two key mechanisms: *Similarity-Aware Inter-domain Compactness* (SIC) and *Domain-Adaptive Generalization* (DAG). The framework consists of four components: (1) an LLM-based item semantic encoding module with lightweight dual projection heads (Section 4.1); (2) a sequential modeling module and final recommendation (Section 4.2); (3) a similarity-aware semantic generalization module for selective inter-domain alignment (Section 4.3); and (4) a domain-adaptive weighting mechanism for discrepancy-aware regularization (Section 4.4).

4.1. Item Semantic Encoding and Dual Projection

Let x_i denote the textual metadata of item v_i , such as its title and description text. We first employ a pretrained large language model as the encoder $E(\cdot)$ to extract the semantic representation of item v_i :

$$\mathbf{e}_i^{\text{sem}} = E(x_i), \quad (9)$$

where $\mathbf{e}_i^{\text{sem}} \in \mathbb{R}^{d_h}$, and d_h is the hidden dimension of the semantic encoder. This step provides transferable semantic priors and helps construct a shared semantic space across domains.

Since the raw LLM embedding space is not directly compatible with the downstream sequential recommender, we introduce two lightweight projection heads to map semantic embeddings into task-specific latent spaces:

$$\mathbf{z}_i^{\text{rec}} = \mathbf{W}_r \mathbf{e}_i^{\text{sem}} + \mathbf{b}_r, \quad (10)$$

$$\mathbf{z}_i^{\text{gen}} = \mathbf{W}_g \mathbf{e}_i^{\text{sem}} + \mathbf{b}_g, \quad (11)$$

where $\mathbf{W}_r, \mathbf{W}_g \in \mathbb{R}^{d \times d_h}$ and $\mathbf{b}_r, \mathbf{b}_g \in \mathbb{R}^d$ are trainable parameters. Here, $\mathbf{z}_i^{\text{rec}} \in \mathbb{R}^d$ is the recommendation-oriented item representation, while $\mathbf{z}_i^{\text{gen}} \in \mathbb{R}^d$ is the generalization-oriented item representation used for cross-domain semantic regularization.

To provide a unified item input for the subsequent sequential recommender, we further fuse the two projected representations as

$$\tilde{\mathbf{z}}_i = \mathbf{W}_f [\mathbf{z}_i^{\text{rec}} \parallel \mathbf{z}_i^{\text{gen}}] + \mathbf{b}_f, \quad (12)$$

where $[\cdot \parallel \cdot]$ denotes vector concatenation, $\mathbf{W}_f \in \mathbb{R}^{d \times 2d}$ and $\mathbf{b}_f \in \mathbb{R}^d$ are trainable parameters, and $\tilde{\mathbf{z}}_i \in \mathbb{R}^d$ is the fused item representation fed into the sequential modeling module.

Since the source and non-source domains have disjoint item ID spaces, we use distinct indices to avoid confusion. Specifically, source-domain items are indexed by i , while non-source-domain items are indexed by j . Accordingly, their fused representations are denoted by $\tilde{\mathbf{z}}_{is}$ and $\tilde{\mathbf{z}}_{jq}$, respectively, where D^q may refer to the target domain or an auxiliary domain. In this way, the semantic encoder serves as a transferable item initializer [32], while the dual projection heads enable joint optimization for recommendation learning and cross-domain generalization.

4.2. Sequential Modeling and Recommendation

Given a source-domain user $u^s \in U^s$ with interaction sequence $\mathbf{s}_{u^s} = [v_{u^s,1}^s, v_{u^s,2}^s, \dots, v_{u^s,T_{u^s}}^s]$, we first replace each interacted item with its fused semantic representation and obtain the source-side input sequence:

$$\tilde{\mathbf{z}}_{u^s} = [\tilde{\mathbf{z}}_{u^s,1}, \tilde{\mathbf{z}}_{u^s,2}, \dots, \tilde{\mathbf{z}}_{u^s,T_{u^s}}], \quad (13)$$

where $\tilde{\mathbf{z}}_{u^s,i}$ denotes the fused representation of item $v_{u^s,i}^s$. The sequence is then encoded by a sequential recommender $\text{SeqRec}(\cdot)$ to capture the user's dynamic preference state:

$$\mathbf{h}_{u^s} = \text{SeqRec}(\tilde{\mathbf{z}}_{u^s}), \quad (14)$$

where $\mathbf{h}_{u^s} \in \mathbb{R}^d$ is the sequence-level user representation. Here, $\text{SeqRec}(\cdot)$ can be implemented using standard sequential recommenders, such as GRU4Rec, SASRec, or BERT4Rec.

Based on the learned sequence representation, the source-domain relevance score for a candidate item $v_i^s \in V^s$ is computed as

$$\text{score}^s(u^s, i) = \mathbf{h}_{u^s}^\top \mathbf{z}_{is}^{\text{rec}}, \quad (15)$$

where $\mathbf{z}_{is}^{\text{rec}}$ denotes the recommendation-oriented projected representation of item v_i^s . This scoring process constitutes the semantic-sequential recommendation backbone in the source domain.

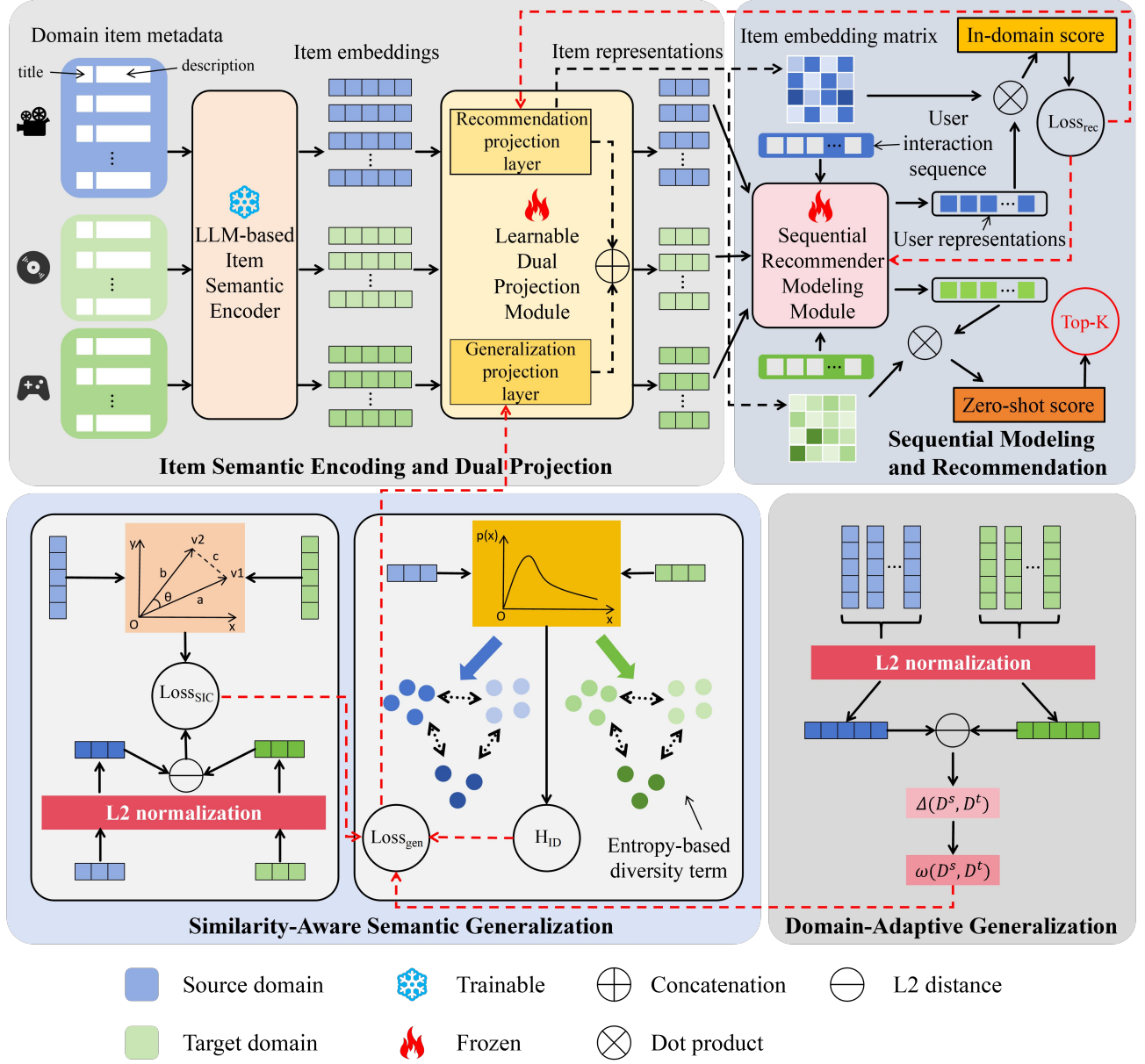


Figure 1: Overall framework of SAGERec.

At inference time, for a target-domain user $u^t \in U^t$ with observed interaction sequence $\mathbf{s}_{u^t} = [v_{u^t,1}^t, v_{u^t,2}^t, \dots, v_{u^t,T_{u^t}}^t]$, we analogously construct the target-side input sequence

$$\tilde{\mathbf{z}}_{u^t} = [\tilde{\mathbf{z}}_{u^t,1}^t, \tilde{\mathbf{z}}_{u^t,2}^t, \dots, \tilde{\mathbf{z}}_{u^t,T_{u^t}}^t], \quad (16)$$

and obtain its sequence representation by

$$\mathbf{h}_{u^t} = \text{SeqRec}(\tilde{\mathbf{z}}_{u^t}). \quad (17)$$

The learned scoring function is then directly transferred to the target domain for zero-shot ranking:

$$\text{score}^t(u^t, j) = \mathbf{h}_{u^t}^\top \mathbf{z}_{j^t}^{\text{rec}}, \quad (18)$$

where $\mathbf{z}_{j^t}^{\text{rec}}$ denotes the recommendation-oriented projected representation of candidate item $v_j^t \in V^t$.

The above module defines the semantic-sequential transfer pipeline of SAGERec, where source-domain interaction sequences are used to learn sequential preference patterns, and the learned scoring function is directly generalized to target-domain next-item recommendation in a zero-shot manner.

4.3. Similarity-Aware Semantic Generalization

A major challenge in zero-shot transfer is that indiscriminate global alignment often collapses the latent space, potentially triggering negative transfer when items from the source domain and non-source domains are only weakly related. Instead of enforcing coarse global attraction, we

introduce a similarity-aware semantic generalization objective, which selectively aligns semantically compatible inter-domain items while preserving discrimination in each domain.

During training, we sample non-source items from D^{ns} for cross-domain semantic regularization. Importantly, only item-side semantic metadata from non-source domains are used, while no interaction data from these domains are involved in optimization. Therefore, the proposed regularization remains compatible with the zero-shot setting.

4.3.1. Similarity-Aware Inter-domain Compactness

Let $v_i^s \in V^s$ and $v_j^q \in V^q$ denote a source-domain item and a sampled non-source item, respectively, where $D^q \in D^{\text{ns}}$. For each source item and sampled non-source item, their semantic embeddings and projected representations are obtained following Section 4.1. We first measure their semantic affinity based on the original LLM embeddings:

$$a_{ij} = \cos(\mathbf{e}_i^{\text{sem}}, \mathbf{e}_j^{\text{sem}}) = \frac{(\mathbf{e}_i^{\text{sem}})^\top \mathbf{e}_j^{\text{sem}}}{\|\mathbf{e}_i^{\text{sem}}\|_2 \|\mathbf{e}_j^{\text{sem}}\|_2}. \quad (19)$$

Since negative cosine values are less informative for positive transfer and may introduce unstable attraction directions, we convert the affinity into a non-negative similarity weight:

$$w_{ij} = \max(0, a_{ij}), \quad (20)$$

where w_{ij} ensures that only semantically compatible pairs contribute positively to the compactness objective.

We further apply L2 normalization to the generalization-oriented projected representations before measuring pairwise distances:

$$\bar{\mathbf{z}}_{is}^{\text{gen}} = \frac{\mathbf{z}_{is}^{\text{gen}}}{\|\mathbf{z}_{is}^{\text{gen}}\|_2}, \quad \bar{\mathbf{z}}_{jq}^{\text{gen}} = \frac{\mathbf{z}_{jq}^{\text{gen}}}{\|\mathbf{z}_{jq}^{\text{gen}}\|_2}. \quad (21)$$

Let $\mathcal{B}^s$ and $\mathcal{B}^q$ denote the source-domain and non-source item sets appearing in the current mini-batch, respectively, and let $\mathcal{P} \subseteq \mathcal{B}^s \times \mathcal{B}^q$ denote the set of sampled cross-domain item pairs. The similarity-aware inter-domain compactness loss is defined as:

$$\mathcal{L}_{\text{SIC}} = \frac{\sum_{(i,j) \in \mathcal{P}} w_{ij} \|\bar{\mathbf{z}}_{is}^{\text{gen}} - \bar{\mathbf{z}}_{jq}^{\text{gen}}\|_2^2}{\sum_{(i,j) \in \mathcal{P}} w_{ij} + \epsilon}, \quad (22)$$

where $\epsilon > 0$ is a small constant for numerical stability.

This formulation has two desirable properties. First, semantically similar source–query item pairs with larger w_{ij} are pulled closer, encouraging consistent and transferable semantics across domains. Second, weakly related or semantically irrelevant pairs receive negligible weights, thereby suppressing noisy attraction caused by excessive uniformity in alignment. Therefore, $\mathcal{L}_{\text{SIC}}$ can be viewed as a selective compactness objective that explicitly considers item-level transfer heterogeneity.

4.3.2. Intra-domain Diversity Preservation

Although inter-domain compactness is beneficial for semantic transfer, overly strong attraction may compress the representation space and weaken item discrimination within each domain. To alleviate such representation collapse, we introduce an intra-domain diversity preservation term.

For each domain $\kappa \in \{s, q\}$, let $\mathcal{B}^{(\kappa)}$ denote the set of indices of items from domain κ . We then define a similarity-based probability distribution $p_{mn}^{(\kappa)}$ over items in the mini-batch $\mathcal{B}^{(\kappa)}$ as:

$$p_{mn}^{(\kappa)} = \frac{\exp(\cos(\mathbf{z}_m^{(\kappa), \text{gen}}, \mathbf{z}_n^{(\kappa), \text{gen}})/\tau_\kappa)}{\sum_{b \in \mathcal{B}^{(\kappa)}} \exp(\cos(\mathbf{z}_m^{(\kappa), \text{gen}}, \mathbf{z}_b^{(\kappa), \text{gen}})/\tau_\kappa)}, \quad (23)$$

where $\tau_\kappa > 0$ is a temperature parameter controlling the smoothness of the distribution, and $\mathbf{z}_m^{(\kappa), \text{gen}}, \mathbf{z}_n^{(\kappa), \text{gen}}$ denote the generalization-oriented projected representations of the items indexed by $m, n \in \mathcal{B}^{(\kappa)}$, respectively. In practice, this distribution is estimated within the current mini-batch for computational efficiency.

Based on this distribution, we define the entropy-based diversity term as

$$\mathcal{H}_{\text{ID}} = - \sum_{\kappa \in \{s, q\}} \frac{1}{|\mathcal{B}^{(\kappa)}|} \sum_{m \in \mathcal{B}^{(\kappa)}} \sum_{n \in \mathcal{B}^{(\kappa)}} p_{mn}^{(\kappa)} \log p_{mn}^{(\kappa)}. \quad (24)$$

A larger $\mathcal{H}_{\text{ID}}$ corresponds to a higher-entropy intra-domain similarity distribution and hence better preservation of intra-domain diversity. This term counterbalances overly aggressive inter-domain attraction and helps maintain fine-grained semantic discrimination.

4.4. Domain-Adaptive Generalization

Besides item-level heterogeneity, zero-shot transfer also exhibits domain-level heterogeneity: different non-source domains may vary significantly in their semantic distance from the source domain. Using a fixed regularization strength for all transfer scenarios is therefore suboptimal. To address this issue, we introduce a domain-adaptive weighting strategy to enhance semantic generalization.

Specifically, for the current mini-batch, we compute the mean semantic embeddings of the source domain and the sampled non-source query domain:

$$\bar{\mathbf{e}}_s^{\text{sem}} = \frac{\sum_{i \in \mathcal{B}^s} \mathbf{e}_i^{\text{sem}}}{\|\sum_{i \in \mathcal{B}^s} \mathbf{e}_i^{\text{sem}}\|_2}, \quad \bar{\mathbf{e}}_q^{\text{sem}} = \frac{\sum_{j \in \mathcal{B}^q} \mathbf{e}_j^{\text{sem}}}{\|\sum_{j \in \mathcal{B}^q} \mathbf{e}_j^{\text{sem}}\|_2}. \quad (25)$$

Moreover, the semantic discrepancy between the two domains is measured by

$$\Delta(D^s, D^q) = \|\bar{\mathbf{e}}_s^{\text{sem}} - \bar{\mathbf{e}}_q^{\text{sem}}\|_2. \quad (26)$$

Based on the discrepancy obtained by Eq. (26), we define a domain-adaptive generalization coefficient as follows:

$$\omega(D^s, D^q) = \lambda \exp(-\gamma \Delta(D^s, D^q)), \quad (27)$$

where $\lambda > 0$ controls the base strength of semantic regularization, and $\gamma > 0$ controls the sensitivity of the decay with respect to domain discrepancy.

This weighting mechanism yields a monotonic relation between semantic proximity and regularization strength. Specifically, when the source domain and the query domain are semantically close, $\Delta(D^s, D^q)$ is small and $\omega(D^s, D^q)$ remains relatively large, imposing more rigorous semantic constraints on domains with high proximity. Conversely, when the semantic gap is large, $\omega(D^s, D^q)$ is reduced, thus encouraging the model to behave more conservatively and avoid over-aligning distant domains. Therefore, by sampling different query domains $D^q \in \mathcal{D}^{ns}$ during training, SAGERec approximates heterogeneous transfer difficulty and dynamically adjusts the regularization strength accordingly.

4.5. Training Objective

To optimize the generalization-oriented projection space for cross-domain semantic alignment and intra-domain diversity preservation, the semantic generalization term is defined as:

$$\mathcal{L}_{\text{gen}} = \mathcal{L}_{\text{SIC}} - \beta \mathcal{H}_{\text{ID}}, \quad (\beta > 0) \quad (28)$$

where β is a hyper-parameter to balance inter-domain selective compactness and intra-domain diversity preservation.

To optimize the model on the source domain, we adopt a standard pairwise ranking loss:

$$\mathcal{L}_{\text{rec}} = - \sum_{(u^s, v^{s,+}, v^{s,-})} \log \sigma(\text{score}^s(u^s, v^{s,+}) - \text{score}^s(u^s, v^{s,-})), \quad (29)$$

where $(u^s, v^{s,+}, v^{s,-})$ denotes a training triplet consisting of user u^s , a positive item $v^{s,+}$, and a sampled negative item $v^{s,-}$, and $\sigma(\cdot)$ is the sigmoid function.

Overall, the training objective is defined as follows:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \omega(D^s, D^q) \mathcal{L}_{\text{gen}}. \quad (30)$$

Overall, SAGERec is a lightweight enhancement to the standard semantic-sequential transfer pipeline, not a replacement for the backbone itself. It combines similarity-aware selective compactness at the item-pair level with domain-adaptive weighting at the domain level, enabling more robust zero-shot cross-domain transfer across diverse semantic gap while maintaining full compatibility with existing sequential recommenders.

5. Experiments

In this section, we conduct comprehensive experiments to evaluate the proposed **SAGERec** framework. Our experiments are designed to answer the following research questions:

- **RQ1:** How effective is SAGERec for zero-shot cross-domain sequential recommendation compared with strong semantic and generalization-based baselines?
- **RQ2:** What are the individual contributions of similarity-aware inter-domain compactness and domain-adaptive generalization?

Table 1

Statistics of the processed datasets after preprocessing.

Dataset	Items	Users	Avg. Seq. Len.
Amazon Movies & TV (AMT)	84,909	179,784	18.6
Amazon CDs & Vinyl (ACV)	20,089	22,170	19.8
Steam	8,156	26,541	19.7

- **RQ3:** How sensitive is SAGERec to its key hyperparameters and weighting strategies?
- **RQ4:** How efficient is SAGERec in terms of training cost and convergence speed compared with representative baselines?
- **RQ5:** How does SAGERec affect cross-domain semantic alignment and representation quality?

5.1. Experimental Setup

5.1.1. Datasets.

We evaluate **SAGERec** on three public real-world datasets from different domains: **Amazon Movies & TV (AMT)**, **Amazon CDs & Vinyl (ACV)**, and **Steam**. Following prior work [33], AMT and ACV are selected from the Amazon Review corpus, while Steam is adopted as a widely used benchmark dataset for sequential recommendation. All three datasets provide sequential user-item interactions together with item-side textual metadata, which makes them suitable for evaluating recommendation models that jointly exploit behavioral signals and semantic item information.

Domain Split. The dataset split follows the zero-shot cross-domain setting considered in this paper. AMT is the largest dataset in terms of both users and items and is therefore used as the source domain with relatively rich interaction signals. ACV and Steam are treated as non-source domains. In each transfer scenario, one of them is used as the evaluation target domain, while the other serves as an auxiliary domain that provides only item-side semantic information for interaction-free regularization. Consequently, the two zero-shot tasks considered in this paper are AMT $\rightarrow$ ACV (with Steam as the auxiliary domain) and AMT $\rightarrow$ Steam (with ACV as the auxiliary domain).

Data Preprocessing. Following prior studies [5, 34, 35, 36], we apply 10-core filtering to remove users and items with fewer than 10 interactions, and discard items without valid textual metadata. The remaining interactions of each user are then sorted chronologically to form behavior sequences. We adopt the leave-one-out protocol, where the last interaction of each user is used for testing, the penultimate one for validation, and the remaining interactions for training. During evaluation, each ground-truth target item is ranked against 100 randomly sampled negative items. Table 1 reports the statistics of the processed datasets.

5.1.2. Baseline Methods.

To comprehensively evaluate the effectiveness of SAGERec, we compare it with seven baseline models under a unified experimental protocol. Following the baseline design of

LLM-RecG [8], we select models that are directly comparable within the same semantic-sequential transfer pipeline, ensuring that performance gains can be clearly attributed to our generalization strategy rather than architectural differences. Specifically, we build our comparisons around three representative sequential backbones and their semantic variants. In addition, we include a recent universal sequence representation method as a strong text-enhanced transferable recommendation baseline.

Sequential Backbones. We adopt three representative sequential recommenders as backbone architectures for constructing semantic transfer baselines. Specifically, **GRU4Rec** [3] represents recurrent sequential modeling, **SASRec** [5] represents unidirectional self-attentive sequential modeling, and **BERT4Rec** [6] represents bidirectional Transformer-based sequence encoding. These backbones cover the major modeling paradigms in sequential recommendation and provide diverse foundations for evaluating the robustness of the proposed framework.

Semantic Variants (-Sem). For the *-Sem* baselines, we construct **GRU4Rec-Sem**, **SASRec-Sem**, and **BERT4Rec-Sem**. These variants incorporate item-side textual metadata through the same LLM-based semantic encoding and projection pipeline, but do not introduce any additional generalization objective. They are used to evaluate the contribution of semantic enhancement alone on different sequential backbones.

Generalization Variants (-RecG). For the *-RecG* baselines, we construct **GRU4Rec-RecG**, **SASRec-RecG**, and **BERT4Rec-RecG**. These variants further introduce the recommendation generalization mechanism on top of the corresponding *-Sem* models, so as to enhance the transferability of learned item representations in the zero-shot target domain. The comparison between each *-Sem* variant and its corresponding *-RecG* variant reveals the contribution of additional generalization-oriented optimization.

Universal Sequence Representation. We also compare with **UniSRec** [37], a strong universal sequence representation learning method that leverages textual semantics to improve recommendation transferability across scenarios. As a competitive text-enhanced recommendation baseline beyond the six backbone-derived variants, UniSRec provides an important external reference in our experiments.

5.1.3. Evaluation Metrics.

We adopt $\text{Recall}@K$ and $\text{NDCG}@K$ as evaluation metrics, with $K \in \{10, 20\}$. $\text{Recall}@K$ measures whether the ground-truth item appears in the top- K recommendation list, while $\text{NDCG}@K$ further evaluates the ranking quality by assigning higher scores to correct items ranked closer to the top positions. Since each test instance contains one positive item and multiple sampled negatives, $\text{Recall}@K$ is equivalent to $\text{Hit Ratio}@K$ in this setting. Higher values of both metrics indicate better recommendation performance.

5.1.4. Implementation Details.

All models are implemented in Python 3.10.19 with PyTorch 2.6.0 and optimized using Adam on a server equipped with an NVIDIA RTX 5880 Ada GPU. For semantic encoding, we employ LLM2Vec [38] based on Llama-3-8B-Instruct [39] to transform item metadata into 4096-dimensional dense vectors, which are then projected into the recommendation latent space. Unless otherwise specified, we set the hidden size to 256, the maximum sequence length to 50, and the dropout rate to 0.2. For Transformer-based backbones, we use 2 layers and 2 attention heads. Models are trained with a batch size of 128 and a learning rate of 1×10^{-4} for at most 100 epochs.

For the newly introduced hyperparameters (λ, γ, β) , we perform grid search on the validation set to select the best values. In each transfer scenario, model selection and early stopping are performed according to the best validation performance on the auxiliary domain, while the final reported results are evaluated on the corresponding target domain. This protocol is designed to avoid using target-domain interaction signals for model selection and thus remains consistent with the zero-shot setting. To improve reproducibility, all methods are evaluated with fixed random seeds, and the final results are reported as the average over three independent runs.

5.1.5. Fair Comparison and Reproducibility.

To ensure a fair comparison, all methods use the same train/validation/test split, the same negative sampling strategy, and the same evaluation metrics. More importantly, all semantic-aware methods, including SAGERec and the baselines, are given access to exactly the same item-side semantic information from the source domain, the target domain, and the auxiliary domain, while no interaction data from the target or auxiliary domains are used for optimization. For baselines that require semantic representations, we further use the same pre-computed semantic embeddings and the same preprocessing pipeline whenever applicable. Therefore, performance differences can be attributed to the proposed similarity-aware compactness and domain-adaptive weighting mechanisms rather than to discrepancies in accessible information, data preprocessing, or training protocol.

5.2. Overall Comparison (RQ1)

Table 2 reports the overall zero-shot recommendation performance when AMT is used as the source domain and ACV and Steam are treated as unseen target domains. To more clearly assess the contribution of the proposed framework itself, the results are organized by sequential backbone, so that each *-SAGE* variant can be directly compared with its corresponding *-Sem* and *-RecG* counterparts under the same architecture. Overall, SAGERec consistently improves performance within all three backbone families, and BERT4Rec-SAGE achieves the best average result among all compared models.

Table 2

Overall zero-shot recommendation performance on three real-world datasets. Results are reported in percentage (%). To evaluate the model-agnostic effectiveness of the proposed framework, variants are grouped by their sequential backbones. Underlined values denote the best-performing variant within each backbone family, while **bold** values denote the best overall results in each column. "Avg." represents the arithmetic mean across all metrics and domains.

Models	ACV				Steam				Avg.
	R@10	N@10	R@20	N@20	R@10	N@10	R@20	N@20	
GRU4Rec-Sem	17.16	8.48	30.32	11.78	16.76	8.65	29.36	11.80	16.79
GRU4Rec-RecG	17.82	8.90	31.21	12.25	18.06	9.12	30.71	12.29	17.55
GRU4Rec-SAGE	<u>19.65</u>	<u>9.85</u>	<u>33.13</u>	<u>13.23</u>	<u>20.46</u>	<u>10.01</u>	<u>30.95</u>	<u>12.64</u>	<u>18.74</u>
SASRec-Sem	38.40	20.65	55.71	25.02	21.54	10.79	35.53	14.30	27.74
SASRec-RecG	42.25	22.61	59.58	26.99	22.75	11.44	37.91	15.23	29.85
SASRec-SAGE	<u>45.50</u>	<u>25.20</u>	<u>62.70</u>	<u>29.54</u>	<u>24.30</u>	12.31	<u>38.10</u>	15.78	<u>31.68</u>
BERT4Rec-Sem	39.74	21.49	56.62	25.74	22.39	11.29	35.94	14.69	28.49
BERT4Rec-RecG	40.65	22.18	57.89	26.52	22.51	11.00	36.34	14.47	28.95
BERT4Rec-SAGE	45.99	25.43	62.93	29.71	24.45	<u>12.15</u>	38.36	<u>15.65</u>	31.83
UniSRec	40.16	21.14	58.14	25.67	20.42	10.15	34.65	13.71	28.01

- (1) *Consistent family-wise gains.* Across all three sequential backbones, the *-SAGE* variant achieves the best performance within its corresponding model family. For the GRU4Rec backbone, the average score improves from 16.79 (*-Sem*) and 17.55 (*-RecG*) to 18.74 (*-SAGE*). For SASRec, the average score further increases from 27.74 and 29.85 to 31.68. A similar trend is observed for BERT4Rec, where *-SAGE* reaches the highest average score of 31.83, compared with 28.49 for *-Sem* and 28.95 for *-RecG*. These results indicate that the gains of SAGERec are not merely due to stronger backbones, but can also be consistently observed within the same backbone architecture.
- (2) *Advantages over uniform generalization.* Comparing each *-SAGE* variant with its corresponding *-RecG* counterpart further suggests that the proposed similarity-aware and discrepancy-adaptive design is more effective than a uniform cross-domain generalization strategy. The advantage is particularly clear on the Transformer-based backbones. For SASRec, the *-SAGE* variant outperforms *-RecG* on both ACV and Steam across the evaluated metrics. The same trend is observed for BERT4Rec, where the gains are especially notable on ACV and remain positive on Steam. On GRU4Rec, the improvement is also consistent, but the absolute margins are relatively smaller than those on SASRec and BERT4Rec. This indicates that selective cross-domain alignment is generally more beneficial than uniform semantic attraction, although the magnitude of the gain still depends on backbone expressiveness.
- (3) *Stronger benefits on expressive backbones.* The benefit of SAGERec becomes more substantial on stronger sequential backbones. In terms of average performance, SAGERec improves over the corresponding *-RecG* baseline by 1.19 points on GRU4Rec, 1.83 points on SASRec, and 2.88 points on BERT4Rec. This pattern suggests that more expressive sequential encoders are better able to exploit the transferable semantic structure introduced by SAGERec. At the same time, the table also shows that the advantage is not equally large in every setting. In particular, the margins within the GRU4Rec family are visibly smaller, indicating that a weaker sequential encoder may limit the extent to which semantic regularization can be translated into recommendation gains.
- (4) *Overall competitiveness against a strong external baseline.* Beyond the within-family comparisons, UniSRec serves as a strong external transferable baseline. The results show that both SASRec-SAGE and BERT4Rec-SAGE outperform UniSRec in terms of average performance, while BERT4Rec-SAGE achieves the best overall average score of 31.83 among all compared models. More specifically, BERT4Rec-SAGE obtains the best results on all four metrics in the AMT→ACV task, as well as the best Recall@10 and Recall@20 in the AMT→Steam task. Meanwhile, SASRec-SAGE achieves the best NDCG@10 and NDCG@20 on Steam, indicating slightly better ranking quality at the top positions in that scenario. Overall, these results support the effectiveness and robustness of SAGERec in zero-shot cross-domain sequential recommendation.

5.3. Ablation Study (RQ2)

To evaluate the contribution of the proposed components, we conduct ablation studies on three representative sequential backbones. Following prior work [8], the intra-domain diversity regularization is retained in all variants to ensure a fair comparison. We compare the full SAGERec model with two degraded variants (*w/o SIC* and *w/o DAG*) as well as the corresponding *-RecG* baseline. The results are summarized in Table 3.

Table 3

Ablation study of SAGERec on different sequential backbones. AMT is used as the source domain, while ACV and Steam are treated as unseen target domains for zero-shot transfer. "Avg." denotes the average of ACV/Steam results on R@10 and N@10. All results are reported in percentage (%).

Backbone	Variant	ACV		Steam		Avg.
		R@10	N@10	R@10	N@10	
GRU4Rec	SAGE	19.65	9.85	20.46	10.01	14.99
	w/o SIC	17.79	8.95	17.02	8.92	13.17
	w/o DAG	16.91	8.42	18.05	9.17	13.14
	RecG	17.82	8.90	18.06	9.12	13.47
SASRec	SAGE	45.50	25.20	24.30	12.31	26.83
	w/o SIC	43.26	23.67	22.20	11.19	25.08
	w/o DAG	44.67	25.02	23.61	11.67	26.24
	RecG	42.25	22.61	22.75	11.44	24.76
BERT4Rec	SAGE	45.99	25.43	24.45	12.15	27.01
	w/o SIC	41.37	22.17	24.23	12.16	24.98
	w/o DAG	43.74	24.03	23.39	12.16	25.83
	RecG	40.65	22.18	22.51	11.00	24.08

Overall, the full SAGERec model achieves the best average (Avg.) performance across all backbones, indicating that the proposed components contribute positively and work in a complementary manner. Two observations can be drawn.

- (1) *Effect of SIC.* Removing the SIC module leads to the most noticeable performance degradation in most cases, especially on the Transformer-based backbones. On BERT4Rec, the average performance drops from 27.01 to 24.98 when SIC is removed, corresponding to a relative decrease of about 7.5%. Similar trends can be observed on SASRec and GRU4Rec, where the Avg. score decreases from 26.83 to 25.08 and from 14.99 to 13.17, respectively. These results suggest that item-level selective alignment plays a central role in effective zero-shot transfer, since it directly filters out weakly related cross-domain correspondences that may otherwise introduce semantic noise.
- (2) *Effect of DAG.* Removing DAG also results in consistent performance drops across different backbones, although the magnitude is generally smaller than that of removing SIC. For example, the Avg. score decreases from 27.01 to 25.83 on BERT4Rec and from 26.83 to 26.24 on SASRec when DAG is removed. On GRU4Rec, the corresponding drop is from 14.99 to 13.14. This indicates that discrepancy-aware weighting further improves transfer robustness by adjusting the strength of semantic regularization according to domain differences, even though its contribution is typically more moderate than that of SIC.

It is also observed that the full SAGERec model does not strictly outperform all degraded variants on every individual metric. For instance, on BERT4Rec, the N@10 result on Steam (12.15) is slightly lower than that of both *w/o SIC* and *w/o DAG* (12.16). However, such minor fluctuations are

negligible compared with the overall average gains across backbones and domains. Overall, the ablation results show that SIC and DAG provide complementary benefits, with SIC serving as the primary source of improvement and DAG acting as a stabilizing factor for heterogeneous cross-domain transfer.

5.4. Sensitivity Analysis (RQ3)

We further examine the sensitivity of SAGERec to three key hyperparameters: the base regularization strength λ , the intra-domain diversity weight β , and the semantic decay rate γ . To isolate the effect of each parameter while controlling computational cost, we adopt a coordinate-wise evaluation strategy on the BERT4Rec-based variant, varying one hyperparameter at a time while fixing the others at their default values. Specifically, we evaluate $\lambda \in \{0.005, 0.010, 0.050\}$, $\beta \in \{0.1, 0.3, 0.7\}$, and $\gamma \in \{0.05, 0.10, 0.20\}$. The results are shown in Figure 2.

Overall, SAGERec shows relatively stable behavior within the tested ranges, while still exhibiting meaningful sensitivity to the three hyperparameters. Several observations can be drawn.

- **Base regularization strength (λ).** Performance improves steadily as λ increases within the tested range, and the best results are obtained at $\lambda = 0.05$ for both Recall@10 and NDCG@10. This trend suggests that a sufficiently strong semantic regularization term is beneficial for reducing cross-domain discrepancy and improving zero-shot transfer.
- **Intra-domain diversity weight (β).** The best performance is obtained at the smallest tested value, $\beta = 0.1$, while larger values lead to a consistent decline in both metrics. This result indicates that intra-domain diversity preservation is helpful, but assigning it an excessively large weight may over-constrain the representation space and reduce the flexibility required for effective cross-domain alignment.
- **Semantic decay rate (γ).** The performance follows a non-monotonic pattern, peaking at $\gamma = 0.10$. When γ is too large, the weighting scheme becomes overly selective and may under-utilize transferable semantic relations from moderately distant domains; conversely, when γ is too small, the selective effect is weakened, and the weighting becomes nearly uniform, which may introduce noisy correspondences from semantically distant domains. This pattern is consistent with the intended role of γ in balancing selectivity and coverage in cross-domain alignment.

5.5. Efficiency Analysis (RQ4)

Training efficiency is important for zero-shot cross-domain sequential recommendation because practical deployment requires strong transferability without excessive optimization cost. To assess the efficiency of SAGERec, we compare *-Sem*, *-RecG*, and *-SAGE* on three backbones, namely GRU4Rec, SASRec, and BERT4Rec. All models are

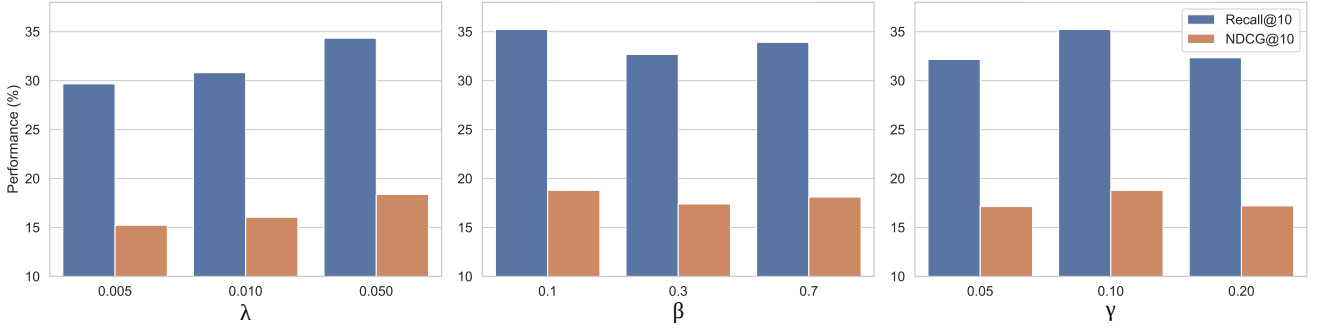


Figure 2: Sensitivity analysis of the key hyperparameters (λ , β , and γ) for BERT4Rec-SAGE. Results are reported in terms of average Recall@10 and average NDCG@10 over the two target domains.

Table 4

Efficiency comparison across sequential backbones. All models are trained on the same device under the same optimization settings; the reported epoch numbers correspond to the stopping point selected by the validation protocol.

Backbone	Variant	Epochs	Total (min)	Avg./epoch (s)
GRU4Rec	Sem	39	27.41	42.17
	RecG	54	107.71	119.68
	SAGE	42	32.55	46.50
SASRec	Sem	60	40.86	40.86
	RecG	27	75.95	168.79
	SAGE	51	39.86	46.89
BERT4Rec	Sem	48	40.81	51.02
	RecG	30	96.69	193.38
	SAGE	30	23.11	46.22

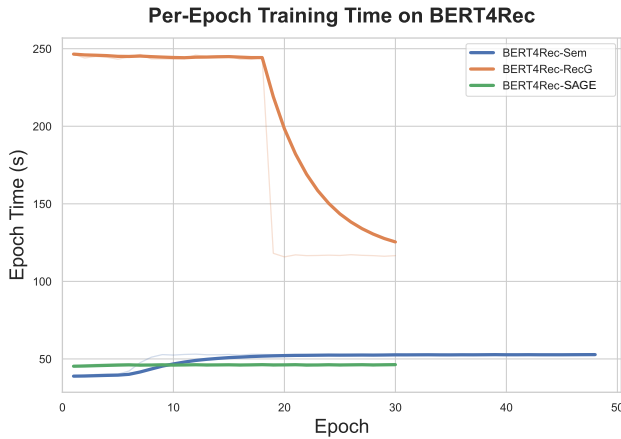


Figure 3: Per-epoch training time on the BERT4Rec backbone. Compared with -RecG, SAGERec maintains a substantially lower and more stable per-epoch overhead throughout training.

trained sequentially on the same device under the same optimization settings. Since the same pre-computed semantic embeddings are shared by all semantic-aware variants in this comparison, the one-off semantic encoding stage is treated as offline preprocessing and excluded from the reported cost.

We report both the average epoch time and the total training time up to the stopping epoch selected by the validation protocol, with the former serving as the primary indicator of optimization overhead and the latter providing an end-to-end reference. As shown in Table 4, SAGERec consistently incurs much lower epoch-level overhead than -RecG across all three backbones. In terms of per-epoch training time, SAGERec achieves about 2.57 $\times$, 3.60 $\times$, and 4.18 $\times$ speedup over -RecG on GRU4Rec, SASRec, and BERT4Rec, respectively. On BERT4Rec, for example, SAGE reduces the average epoch time from 193.38 to 46.22 seconds, and the total training time from 96.69 to 23.11 minutes. Similar reductions are also observed on SASRec and GRU4Rec, indicating that the proposed selective alignment strategy is substantially lighter than the denser RecG-style generalization objective. A likely reason is that -RecG relies on denser cross-domain alignment, whereas SAGERec reduces unnecessary alignment computations through similarity-aware selection.

Figure 3 further illustrates the epoch-wise training cost on the representative BERT4Rec backbone. -RecG exhibits substantially higher and less stable epoch-level overhead throughout training, whereas SAGERec remains comparatively low and stable. Compared with the lightweight -Sem baseline, SAGERec introduces only marginal additional overhead while remaining markedly more efficient than -RecG. Notably, on SASRec, although SAGERec requires more training epochs than -RecG (51 vs. 27), its much lower per-epoch cost still leads to a substantially shorter total training time (39.86 vs. 75.95 minutes). These results suggest that the SIC and DAG modules provide a favorable efficiency–effectiveness trade-off, allowing SAGERec to improve zero-shot recommendation performance without incurring prohibitive training overhead.

5.6. Representation Analysis (RQ5)

To further understand how SAGERec influences zero-shot transfer, we visualize both the learned item representations and the resulting cross-domain alignment patterns. Figure 4 provides two complementary views of the learned representation space.

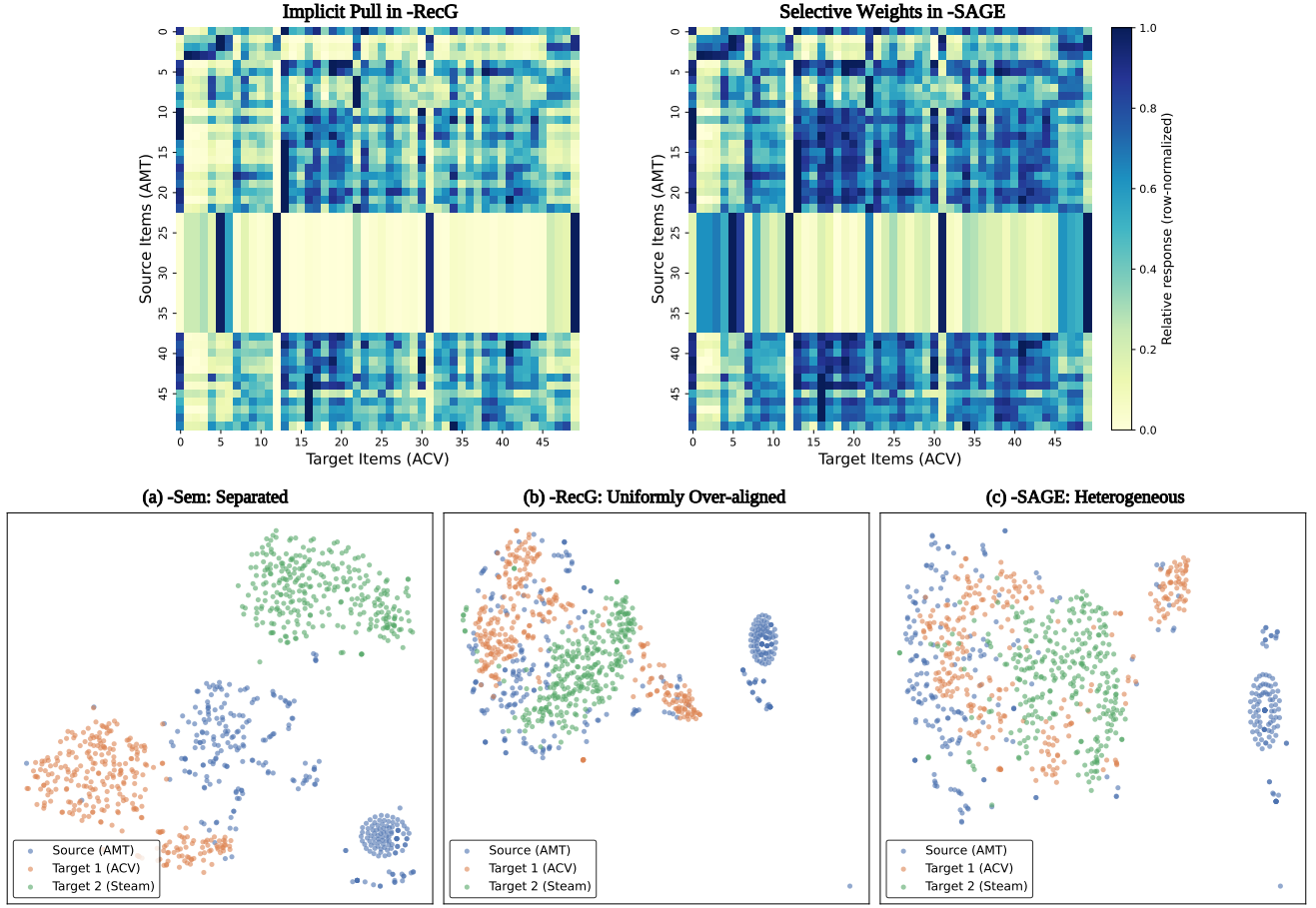


Figure 4: Representation analysis of SAGERec. Top: row-normalized cross-domain alignment weight matrices between 50 sampled source items and 50 target items in the AMT→ACV setting, displayed under a shared color scale for structural comparison. Bottom: t-SNE visualization of projected item representations from the BERT4Rec-based variants across the source domain (AMT) and two non-source domains (ACV and Steam).

- (1) *Item-level cross-domain weighting patterns.* The top part of Figure 4 visualizes the row-normalized cross-domain weight matrices between 50 sampled source items and 50 sampled target items in the AMT→ACV setting. Under a shared color scale, the -RecG variant exhibits a comparatively diffuse and less differentiated response pattern, suggesting limited selectivity when aligning cross-domain item pairs. By contrast, SAGERec shows a more structured and contrastive distribution, with clearer bands and localized high-response regions over cross-domain correspondences. This observation is consistent with the intended role of the SIC module, namely, to prioritize reliable transfer signals while suppressing noisier alignments.
- (2) *Domain-level alignment in the latent space.* In the bottom part of Figure 4, we project the learned item representations of the BERT4Rec-based variants into a two-dimensional space using t-SNE. The -Sem variant shows relatively separated clusters across the source and non-source domains, suggesting that semantic enhancement alone is insufficient to establish

effective cross-domain alignment. The -RecG variant produces substantially stronger overlap among the three domains, but the projected structure also appears more mixed and less domain-distinguishable. In contrast, SAGERec yields a more structured arrangement: some cross-domain overlap is retained to support transfer, while clearer domain-specific discrimination is still preserved. This pattern is consistent with the design goal of achieving selective rather than uniform alignment.

Overall, the visualization results are qualitatively consistent with the quantitative improvements reported in Table 2. They suggest that SAGERec better balances cross-domain alignment and intra-domain discrimination than semantic enhancement alone or uniform generalization.

5.7. Discussion

The results from RQ1–RQ5 collectively suggest that transfer heterogeneity plays an important role in zero-shot cross-domain sequential recommendation. Across different sequential backbones, SAGERec improves recommendation

performance over semantic-only and uniform-generalization baselines while maintaining relatively low optimization overhead. These findings indicate that explicitly modeling cross-domain heterogeneity is beneficial when source and target domains differ substantially in semantics.

A key observation is that uniform cross-domain alignment is not always appropriate for the zero-shot transfer setting. As evidenced by the ablation and representation analyses, non-selective attraction across all cross-domain item pairs can weaken intra-domain structure and introduce noisy correspondences. From this perspective, the SIC module addresses a more fundamental challenge at the item level than the domain-level weighting issue handled by DAG, as it explicitly prioritizes semantically compatible pairs instead of enforcing indiscriminate alignment.

The results also indicate that regularization strength should vary with domain discrepancy. The performance differences across target domains and backbones, together with the sensitivity analysis, suggest that a fixed regularization scheme is insufficient for heterogeneous transfer scenarios. In this regard, the DAG mechanism acts as a discrepancy-aware regulator that adjusts semantic regularization according to domain distance, making transfer more conservative when cross-domain similarity is weak.

At the same time, the current results reveal the limitation of relying mainly on item-side semantic transfer. Although LLM-derived embeddings provide a useful bridge across domains, some cross-domain gaps remain difficult to resolve using item metadata alone, especially when domain-specific user intent or platform-dependent semantics play a dominant role. This limitation may explain why the gains are smaller in more distant transfer scenarios, and suggests that future work should consider finer-grained modeling of domain-specific bias and more adaptive user-level transfer mechanisms.

6. Conclusion

In this paper, we study zero-shot cross-domain sequential recommendation from the perspective of transfer heterogeneity. We identify a key limitation of existing semantic transfer methods, namely, their tendency to impose overly uniform cross-domain alignment, which may introduce noisy correspondences and weaken intra-domain structure. To address this issue, we propose SAGERec, a similarity-aware generalization framework that improves zero-shot transfer through two components: the Similarity-Aware Inter-domain Compactness (SIC) module for selective item-level alignment, and the Domain-Adaptive Generalization (DAG) mechanism for discrepancy-aware regularization.

Extensive experiments on real-world datasets and multiple sequential backbones show that SAGERec consistently improves recommendation performance within matched backbone families and achieves strong overall results across transfer settings. The ablation, sensitivity, efficiency, and representation analyses further indicate that the proposed

framework provides an effective balance between cross-domain alignment, intra-domain discrimination, and optimization cost, highlighting the importance of modeling transfer heterogeneity in zero-shot cross-domain sequential recommendation.

Despite these encouraging results, the current framework still relies primarily on item-side semantic transfer, which may be insufficient for handling more complex cross-domain gaps such as domain-specific user intent and platform-dependent semantics. Future work may therefore explore finer-grained modeling of domain-specific semantic bias and more adaptive mechanisms for user-level transfer in highly heterogeneous recommendation scenarios.

CRedit authorship contribution statement

Zihao Pan: Writing – original draft, Methodology, Conceptualization, Validation. **Yuzhuo Dang:** Validation, Methodology, Formal analysis. **Xin Zhang:** Formal analysis, Conceptualization. **Zhiqiang Pan:** Investigation, Project administration. **Taihua Shao:** Writing – review & editing, Investigation. **Fei Cai:** Resources, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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